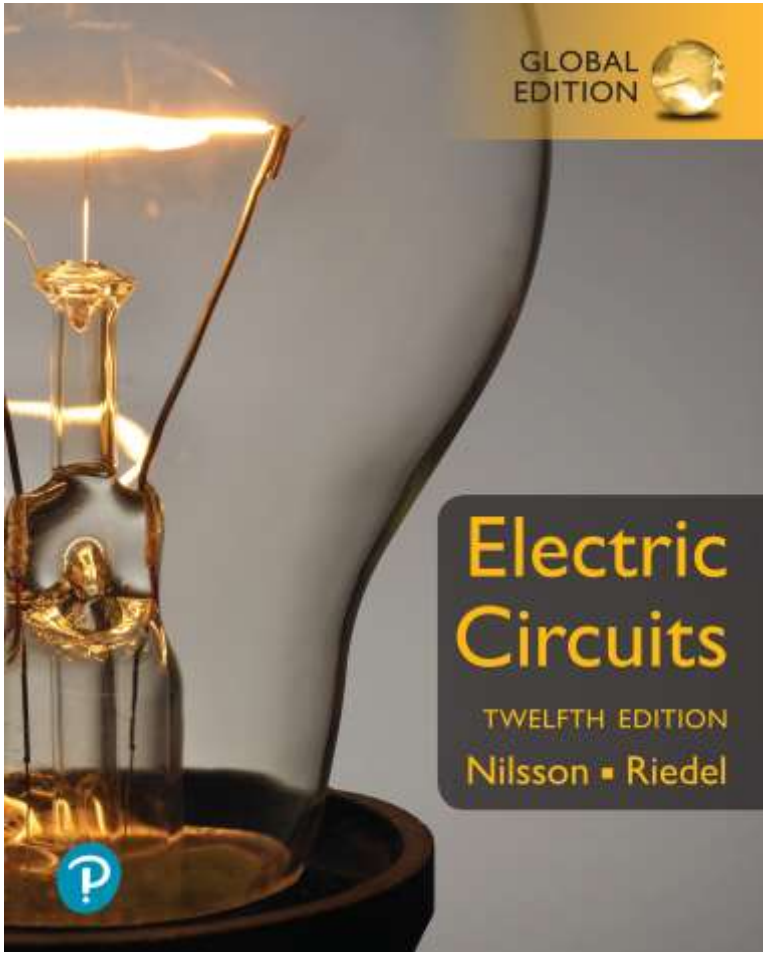
Electric Circuits

Twelfth Edition, Global Edition



Chapter 6 – Part 1

Inductance, Capacitance, and Mutual Inductance

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Chapter 6 – Inductance and Capacitance

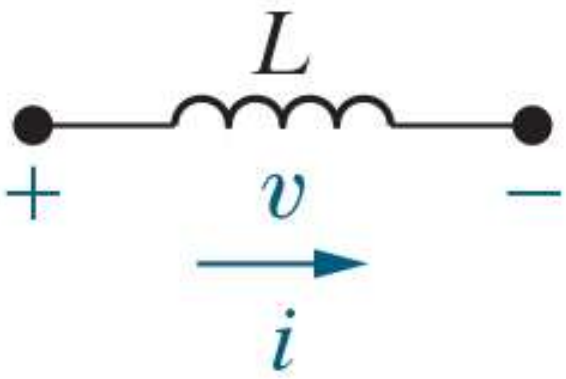
Objectives:

- **6.1** Inductor
  - **6.1.1** Equations for v, i, p, and w
  - **6.1.2** Behavior in the presence of constant current
  - **6.1.3** Continuity requirements
  - **6.1.4** Combine in series and in parallel
- **6.2** Capacitor
  - **6.2.1** Equations for v, i, p, and w
  - **6.2.2** Behavior in the presence of constant voltage
  - **6.2.3** Continuity requirements
  - **6.2.4** Combine in series and in parallel

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Chapter 6 – Inductors (1 of 17)



- Coil of wire with time-varying current
  - time-varying magnetic field
  - time-varying voltage drop
- Inductor equation:

$$v(t) = L\frac{di(t)}{dt}$$

- Units: v(t) is volts, i(t) is amps, and L is henries [H]

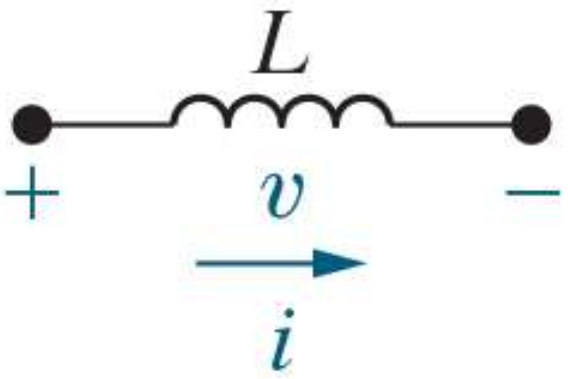
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Chapter 6 – Inductors (2 of 17)

Look at the inductor equation again:

$$v(t) = L\frac{di(t)}{dt}$$



Suppose i(t) is constant. Then v(t) =

A. L

B. 0

C. Undefined

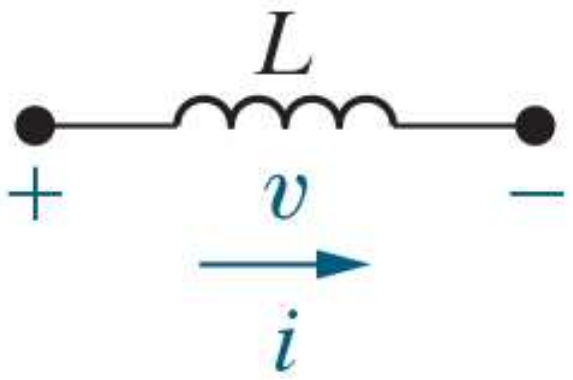
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Chapter 6 – Inductors (2 of 17)

Look at the inductor equation again:

$$v(t) = L\frac{di(t)}{dt}$$



Suppose i(t) is constant. Then v(t) =

- ✗ A. L
- ✓ B. 0
- ✗ C. Undefined

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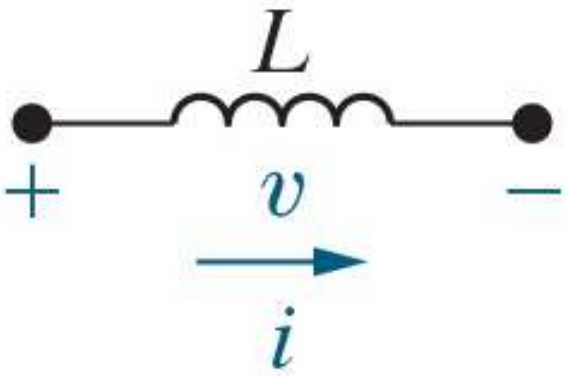
Chapter 6 – Inductors (3 of 17)

So, if the current in an inductor is constant, its voltage drop is 0, so the inductor can be replaced by

A. A short circuit

B. An open circuit

C. A capacitor



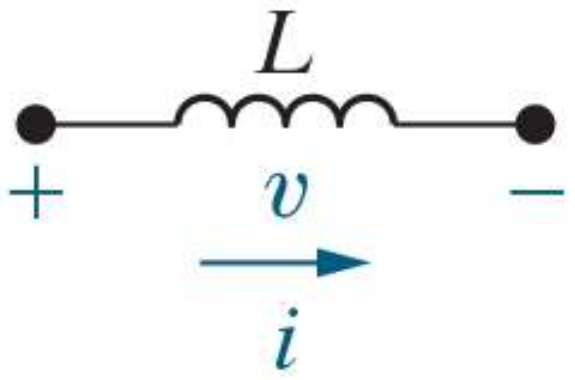
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Chapter 6 – Inductors (3 of 17)

So, if the current in an inductor is constant, its voltage drop is 0, so the inductor can be replaced by

- ✔ A. A short circuit
- ✘ B. An open circuit
- ✘ C. A capacitor



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Chapter 6 – Inductors (6 of 17)

- The equation for current in terms of voltage:

$$v(t) = L\frac{di(t)}{dt} \quad \Rightarrow \quad v(t)dt = Ldi(t)$$

$$\Rightarrow \quad \int_{t_o}^{t} v(\tau)d\tau = L\int_{i(t_o)}^{i(t)} dx$$

$$\Rightarrow \quad i(t) = \frac{1}{L}\int_{t_o}^{t} v(\tau)d\tau + i(t_o)$$

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Chapter 6 – Inductors (7 of 17)

- Power and energy

$$p(t) = v(t)i(t) = Li(t)\frac{di(t)}{dt}$$

$$p(t) = \frac{dw(t)}{dt} = Li(t)\frac{di(t)}{dt}$$

$$\Rightarrow \qquad dw(\tau) = Li(\tau)di(\tau)$$

$$\Rightarrow \qquad \int_0^{w(t)} dx = L\int_0^{i(t)} y(\tau)d\tau$$

$$\Rightarrow \qquad w(t) = \tfrac{1}{2}Li(t)^2$$

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Chapter 6 – Inductors (8 of 17)

- Inductors are **energy storage devices**!
  - If the initial current through the inductor is non-zero, the inductor is storing energy.

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Chapter 6 – Inductors (9 of 17)

Suppose the initial current through a 2 mH inductor is 1 A. The initial energy stored in the inductor is

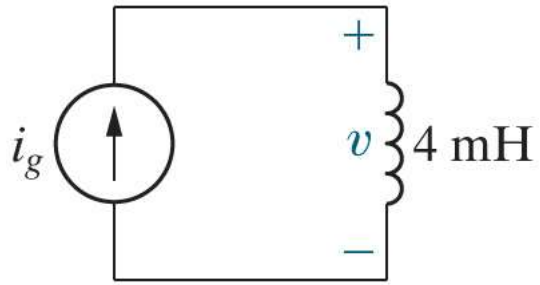
- ✔ A. 1 mJ
- ✘ B. 2 mJ
- ✘ C. 4 mJ

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Chapter 6 – Inductors (10 of 17)

- Example



$$i_g(t) = 0, \qquad t < 0,$$

$$i_g(t) = 8e^{-300t} - 8e^{-1200t} \text{ A}, \qquad t \geq 0.$$

$$v(t) = 0, \qquad t < 0$$

$$v(t) = L\frac{di(t)}{dt} = (0.004)[(-300)8e^{-300t} - (-1200)8e^{-1200t}]$$

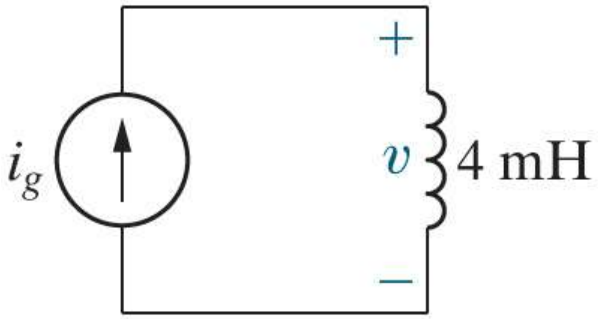
$$= -9.6e^{-300t} + 38.4e^{-1200t} \text{ V}, \qquad t \geq 0$$

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Chapter 6 – Inductors (11 of 17)

- Example (continued)



$$i_g(t) = 0, \qquad t < 0,$$

$$i_g(t) = 8e^{-300t} - 8e^{-1200t} \text{ A}, \qquad t \geq 0.$$

Find the power for the inductor:

$$p(t) = v(t)i(t)$$

$$= (-9.6e^{-300t} + 38.4e^{-1200t})(8e^{-300t} - 8e^{-1200t})$$

$$= -76.8e^{-600t} + 384e^{-1500t} - 307.2e^{-2400t} \text{ W}$$

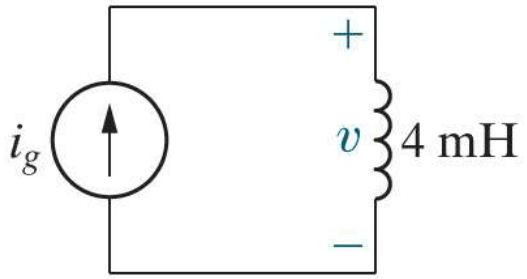
To find the max power and the time at which the power is max, take the first derivative of p(t) and set it equal to 0.

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Chapter 6 – Inductors (12 of 17)

- Example (continued)



$$i_g(t) = 0, \qquad t < 0,$$

$$i_g(t) = 8e^{-300t} - 8e^{-1200t} \text{ A}, \qquad t \geq 0.$$

Find the energy for the inductor:

$$w(t) = \tfrac{1}{2}Li(t)^2$$

$$= \tfrac{1}{2}(0.004)(8e^{-300t} - 8e^{-1200t})^2$$

$$= 128(e^{-600t} - 2e^{-1500t} + e^{-2400t}) \text{ mJ}$$

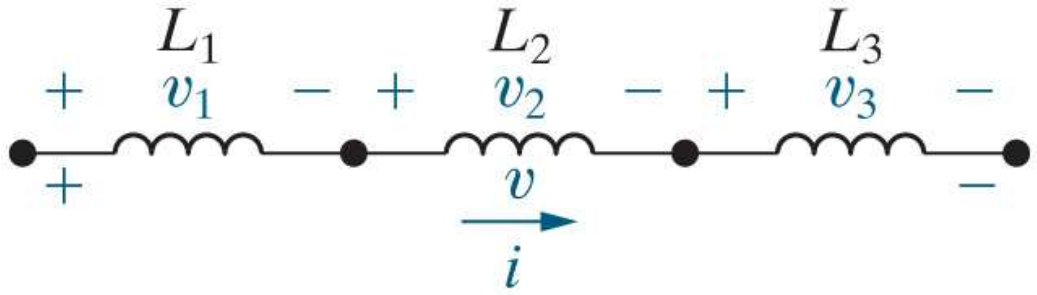
To find the max energy and the time at which the energy is max, take the first derivative of w(t) and set it equal to 0. Or if you don't have w(t) yet, do the same for i(t)!

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Chapter 6 – Inductors (13 of 17)

Inductors connected in series:



$$v_1 = L_1 \frac{di}{dt}, \qquad v_2 = L_2 \frac{di}{dt}, \qquad \text{and} \qquad v_3 = L_3 \frac{di}{dt}.$$

KVL:
$$v = v_1 + v_2 + v_3 = (L_1 + L_2 + L_3)\frac{di}{dt},$$

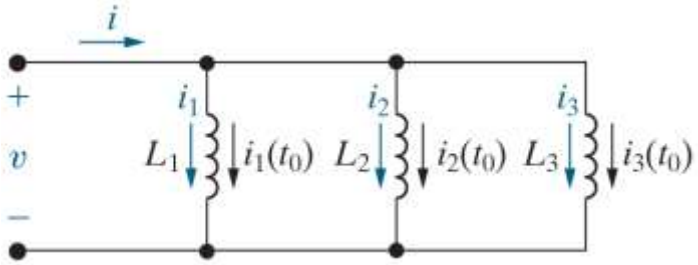
Inductors in series ADD.

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Chapter 6 – Inductors (14 of 17)

Inductors connected in parallel:



$$i_1 = \frac{1}{L_1}\int_{t_0}^{t} v\,d\tau + i_1(t_0), \quad i_2 = \frac{1}{L_2}\int_{t_0}^{t} v\,d\tau + i_2(t_0), \quad i_3 = \frac{1}{L_3}\int_{t_0}^{t} v\,d\tau + i_3(t_0).$$

KCL: $i = i_1 + i_2 + i_3.$

$$i = \left(\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}\right)\int_{t_0}^{t} v\,d\tau + i_1(t_0) + i_2(t_0) + i_3(t_0).$$

$$i = \frac{1}{L_{eq}}\int_{t_0}^{t} v\,d\tau + i(t_0).$$

Inductors combine in parallel like resistors combine in parallel.

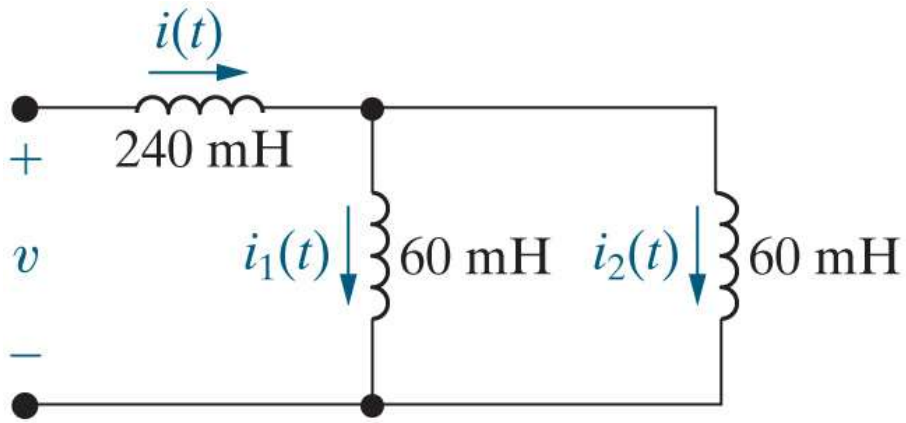
$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} \qquad i(t_0) = i_1(t_0) + i_2(t_0) + i_3(t_0)$$

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Chapter 6 – Inductors (15 of 17)

Find the equivalent inductance for the circuit below, assuming $i_1 = 6\,\text{A}$ and $i_2 = -3\,\text{A}$.



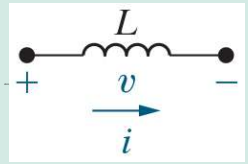
- ✗ A. $L_{eq} = 300\ \text{mH}, i(t) = 6\ \text{A}$
- ✗ B. $L_{eq} = 270\ \text{mH}, i(t) = -3\ \text{A}$
- ✓ C. $L_{eq} = 270\ \text{mH}, i(t) = 3\ \text{A}$

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Chapter 6 – Inductors (16 of 17)

Inductor summary

Equation Type	Equation
Symbol	
Units	Henries [H]
Describing equation	$v(t) = L\dfrac{di(t)}{dt}$
Other equation	$i(t) = \dfrac{1}{L}\int_{t_o}^{t} v(\tau)d\tau + i(t_o)$
Initial condition	$i(t_o)$
Behavior with const. source	If $i(t) = I, v(t) = 0 \rightarrow$ short circuit
Continuity requirement	$i(t)$ is continuous so $v(t)$ is finite

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Chapter 6 – Inductors (17 of 17)

Inductor summary

Equation Type	Equation
Power	$p(t) = v(t)i(t) = Li(t)\dfrac{di(t)}{dt}$
Energy	$w(t) = \frac{1}{2}Li(t)^2$
Initial energy	$w_o(t) = \frac{1}{2}Li(t_o)^2$
Trapped energy	$w(\infty) = \frac{1}{2}Li(\infty)^2$
Series-connected	$L_{eq} = L_1 + L_2 + L_2$ $i_{eq}(t_0) = i(t_0)$
Parallel-connected	$\dfrac{1}{L_{eq}} = \dfrac{1}{L_1} + \dfrac{1}{L_2} + \dfrac{1}{L_3}$ $i_{eq}(t_o) = i_1(t_o) + i_2(t_o) + i_3(t_o)$

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